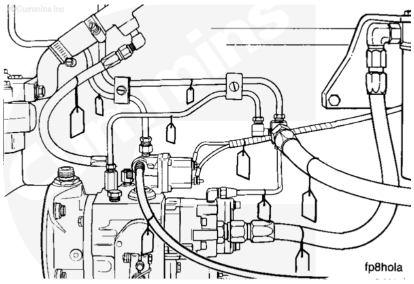
Trainings

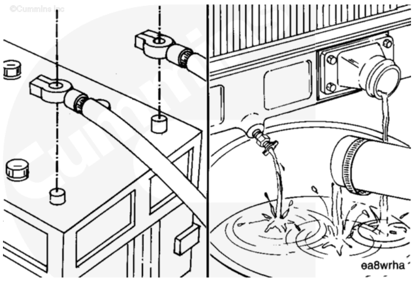
Remove

Tag all hoses, lines, linkage, and electrical connections as removed to identify location and aid in the installation process.



⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

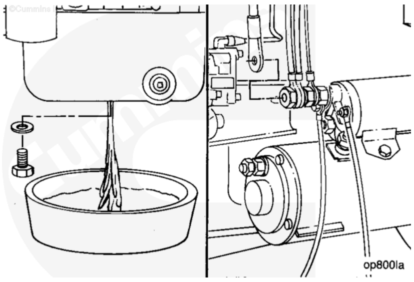
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Disconnect the battery cables. Refer to Procedure 013-009 (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html).

Drain the cooling system. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/89/89-008-018.html).

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.



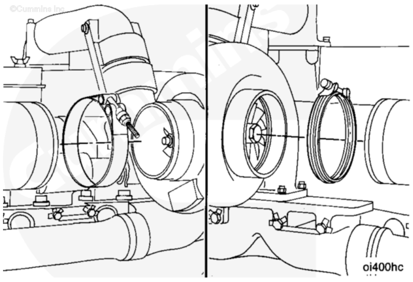
⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

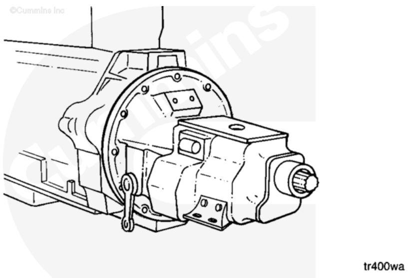
Drain the lubricating oil from the oil pan. Refer to Procedure 007-037 (/qs3/pubsys2/xml/en/procedures/89/89-007-037.html).

Disconnect the starter cable, engine ground straps, cab or chassis to engine hoses, tubing, electrical wiring and hydraulic lines. See equipment manufacturer service information.

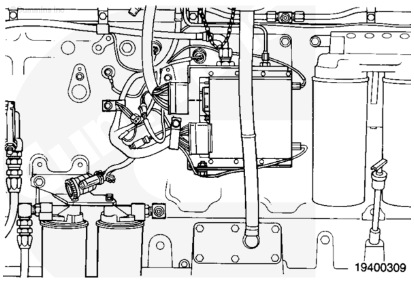
Disconnect the intake and exhaust system piping. Refer to Procedure 010-023 (/qs3/pubsys2/xml/en/procedures/89/89-010-023.html) and Refer to Procedure 011-007 (/qs3/pubsys2/xml/en/procedures/89/89-011-007.html).



Disconnect the drive units from the flywheel housing and flywheel. See equipment manufacturer service information.

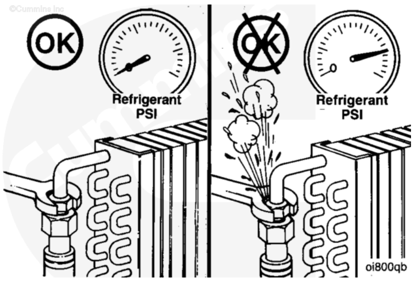


Disconnect the OEM wiring harness from the engine control module (ECM). Refer to Procedure 019-072 (/qs3/pubsys2/xml/en/procedures/19/19-019-072.html) in the Troubleshooting and Repair Manual, Electronic Control System, QSK19, QSK23, QSK45, QSK60, and QSK78 Engines, Bulletin 3666113.



⚠ WARNING ⚠

If a liquid refrigerant system (air conditioning) is used, wear eye and face protection, and wrap a cloth around the fitting before removing. Liquid refrigerant can cause serious eye and skin injuries.

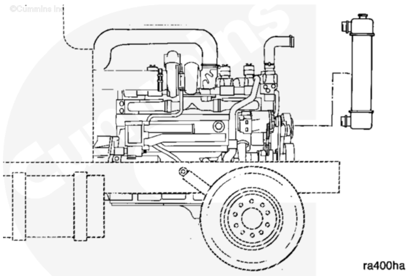


⚠ WARNING ⚠

To protect the environment, liquid refrigerant systems must be properly emptied and filled using equipment that prevents the release of refrigerant gas into the atmosphere. Federal law requires capturing and recycling the refrigerant.

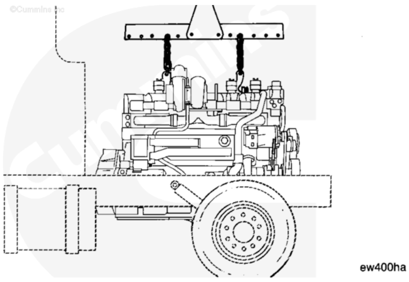
Disconnect the refrigerant lines from the engine. See equipment manufacturer service information.

Remove all chassis components necessary to remove the engine from the equipment. See equipment manufacturer service information.



⚠ WARNING ⚠

The engine lifting equipment must be designed to lift the engine and transmission as an assembly without causing personal injury.



⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

On applications where the rear engine mounts are attached to the drive unit, it is sometimes necessary to remove the engine and drive unit as an assembly.

Inspect the engine lifting brackets for damage or cracks. Do **not** attempt to lift engine if cracks or damage is visible.

Use engine lifting fixture, Part Number 3163264, or equivalent, for removal of engine. The fixture **must** be designed to lift a maximum of 3175 kg [7000 lb]. Refer to the manufacturer's instructions.

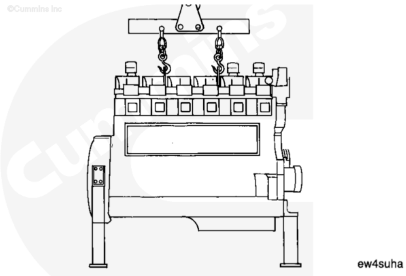
Use a properly rated hoist and engine lifting fixture to support the weight of the engine. Remove the front and rear engine support capscrews.

Use a properly rated hoist, engine lifting fixture, Part Number 3163264, or equivalent, to remove the engine from the chassis.

If the drive unit is **not** removed, place a support under the drive unit to prevent it from falling.

⚠ CAUTION ⚠

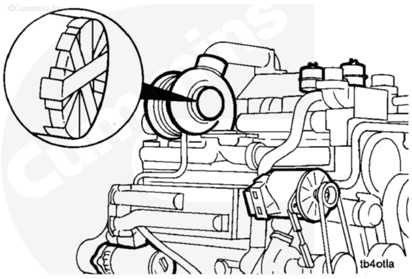
The oil sump or cover plate will not support the weight of the engine. The engine will fall and damage to the engine will occur.



Use a engine stand or skid that contacts the engine mounts. The stand **must** hold the weight of the engine and provide permanent support to prevent the engine from falling.

Place the engine on an engine stand.

Cover all engine openings to prevent dirt and debris from entering the engine.



Remove all remaining accessories and brackets that will be used on the replacement engine.